Name: ______________________

Worksheet

Checking and securing solving linear equations

Task A: Solving equations involving a bracketed expression

1) Spot the mistake in each method to solve an equation involving brackets.

a)

$$\begin{aligned} 5(x+10) &= 55 \\ 5x+10 &= 55 \\ 5x &= 45 \\ x &= 9 \end{aligned}$$

b)

$$\begin{aligned} 2(4x+10) &= 18 \\ 2x+5 &= 9 \\ 2x &= 4 \\ x &= 2 \end{aligned}$$

c)

$$\begin{aligned} 7(x-3) &= 42 \\ x-3 &= 6 \\ x &= 3 \end{aligned}$$

d)

$$\begin{aligned} 5(3x+2) &= 14 \\ 8x+10 &= 14 \\ 8x &= 4 \\ x &= 2 \end{aligned}$$

e)

$$\begin{aligned} \frac{1}{3}(2x-5) &= 15 \\ 2x-5 &= 3 \\ 2x &= 8 \\ x &= 4 \end{aligned}$$

f)

$$\begin{aligned} \frac{3}{4}(10-8x) &= 6 \\ \frac{30}{4}-\frac{24}{4}x &= 6 \\ -\frac{24}{4}x &= -\frac{3}{2} \\ x &= -\frac{3}{2}\times\frac{4}{24} \\ x &= -\frac{1}{4} \end{aligned}$$

2a) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$6(x-9) = 21$$

Multiplying by the reciprocal first:

$$6(x-9) = 21$$

2b) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$15(4x+7) = 45$$

Multiplying by the reciprocal first:

$$15(4x+7) = 45$$

2c) Solve this equation using both methods and write a sentence comparing the efficiency of the methods.

Expanding first:

$$\frac{6}{5}(3x + 4) = 30$$

Multiplying by the reciprocal first:

$$\frac{6}{5}(3x + 4) = 30$$

Task B: Solving equations with multiple brackets

1) The solutions to the equations are given. Fill in the steps to show how to get to these answers. *Other methods may also be valid.*

a)

$5(x + 6) = 7(9x - 4)$

$x = 1$

b)

$2(4x - 5) = -6(3x - 7)$

$x = 2$

c)

$8(x - 5) + 3(x + 5) = 8$

$x = 3$

d)

$4(3x - 1) - 2(3x - 1) = 22$

$x = 4$

e)

$2(4x - 9) - 4(3 - 2x) = 50$

$x = 5$

f)

$3(2(x + 4)+5) = 75$

$x = 6$

2) Solve these equations twice. Firstly by expanding first, secondly by simplifying before expanding. Write a sentence to compare the efficiency.

a) $4(x + 5) + 3(x + 5) = 14$

$4(x + 5) + 3(x + 5) = 14$

b) $21 + 3(2x - 3) = 7(2x - 3) - 3(2x - 3)$

$21 + 3(2x - 3) = 7(2x - 3) - 3(2x - 3)$